School of Engineering and Applied Science (SEAS), Ahmedabad University

CSE 400: Fundamentals of Probability in Computing

Project Scribe Report: Milestone 3

Deterministic Baseline Simulation & Performance Analysis

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Question-1: System Objective and Deterministic Baseline

Clearly restate your project objective. Identify and describe the deterministic baseline strategy you are implementing. Explain why this baseline is reasonable or commonly used in your domain.

Solution:

1. The Probabilistic Problem and System Objective:

The core problem addressed is the **stochastic latency violation** in Device-to-Device (D2D) networks. Because task processing times (X_{proc}) and inter-arrival times (T_{inter}) are random variables, a system based on fixed average values will inevitably fail to meet strict deadlines. Our primary objective is to maximize the **Global Task Completion Rate (TCR)**, defined as:

$$\text{maximize } \mathbb{E}\left[\sum_{i \in N} Y_i\right]$$

where Y_i is a Bernoulli variable that equals 1 if the sojourn time meets the deadline τ, and 0 otherwise.

2. Primary Sources of Uncertainty:

Our simulation explicitly models uncertainty through:

- **Arrival Jitter:** Modeled as a Poisson stream where task arrivals are unpredictable.
- **Workload Volatility:** Modeled via Gamma and Exponential distributions to simulate varying application complexity.
- **Channel Fading:** Distance-based bandwidth degradation adding uncertainty to transmission times (t_{trans}).

3. Deterministic Baseline Strategy: Shortest Delay First (SDF)

For Milestone 3, we implemented a non-randomized, greedy logic where each node evaluates the instantaneous state of its neighbors. The decision is calculated as follows:

- **Local Delay Estimation:** $D_{local} = W_{q,source} + \frac{C_k}{f_{source}}$
- **Remote Delay Estimation:** $D_{remote,j} = t_{trans} + W_{q,j} + \frac{C_k}{f_j}$

The system routes the task to the device $j \in \{source, neighbor\}$ that yields the absolute $\min(D_j)$. To maintain queue stability, we apply a $1.2\times$ **saturation penalty** if a neighbor's queue backlog exceeds 2.0 seconds.

4. Justification for the Baseline:
This "Greedy" approach is a standard industry benchmark in distributed computing. It serves as a perfect scientific control because:

- It isolates the **Congestion Externality**, showing how individual greedy decisions lead to the "Herd Effect."
- It provides a direct performance floor against which we will measure the "Smart" Gain of randomized Game Theoretic algorithms in Milestone 4.

Question-2: Key Random Variables and Uncertainty Modeling

Identify the key random variables in your project and describe how uncertainty is modeled for each. Clearly state any probabilistic assumptions currently being made.

Solution:

Our simulation models the D2D environment as a high-fidelity stochastic system. While the decision logic is deterministic, the inputs are modeled as rigorous random variables (RVs). The following RVs define the uncertainty in our system:

1. Inter-arrival Time (T_{inter}):
Definition: The time interval between task generation events.
Distribution: $T_{inter} \sim$ Exponential(λ), where λ is swept from 0.1 to 1.4 in our validation phase.
Role: This RV drives the Poisson arrival process, creating the unpredictable traffic bursts visualized in our Stage 2 "System Reliability" plot.

2. Task CPU Cycles (C_k):
Definition: The raw computation requirement of a task.
Modeling: We implemented a bimodal workload profile to simulate real-world application diversity (see "Workload Mix" Pie Chart in Stage 2 Dashboard):

- **AR Tasks:** $C_k \sim$ Gamma($k = 2, \theta = 1.5$). This models a General distribution ($M/G/1$) with high variance, capturing heavy-tailed processing risks.
- **BeautyCam Tasks:** $C_k \sim$ Exponential($\mu = 1.0$), following the standard Markovian model ($M/M/1$).

3. Task Payload Size (S_k):
Definition: The amount of data transmitted during an offloading event.
Distribution: $S_{AR} \sim U(0.1, 0.4)$ and $S_{BC} \sim U(2.0, 4.0)$ Megabits.
Role: This RV introduces variance in the transmission delay (t_{trans}), which the deterministic policy must account for when estimating remote delay.

4. Physical Distance ($d_{i,j}$):
Definition: The spatial distance between the source and a neighbor device.
Distribution: $d \sim$ Uniform$(5, 50)$ meters.
Role: Distance is used to calculate effective bandwidth $BW_{eff} = \frac{BW_{base}}{1+(d/20)^2}$. This captures physical-layer uncertainty and channel fading, which significantly influences the routing decision.

5. Queue Backlog (W_q):
Definition: The cumulative processing time of tasks currently waiting in the buffer.
Role: This is a **Derived Random Variable** that evolves based on past realizations of T_{inter} and C_k. It is the primary input for our deterministic Shortest Delay First (SDF) policy.

6. Probabilistic Assumptions:
To ensure our baseline results are scientifically interpretable, we made the following assumptions:

- **Poisson Memoryless Property:** We assume arrivals are independent, meaning the arrival of one task provides no information regarding the timing of the next.

- **Stationary Distributions:** We assume the parameters $(\lambda, \mu, \alpha, \beta)$ remain constant during a traffic burst.
- **i.i.d. Processing:** Task cycle requirements are independent and identically distributed for each application type.

Question-3: Probabilistic Reasoning and Dependencies

Explain how probabilistic relationships (e.g., dependence, independence, conditionality) are used within your project system to support reasoning, inference, or decision-making.

Solution:

To establish our baseline, we implemented a strictly deterministic decision rule that operates within a high-entropy stochastic environment. Our simulation leverages three fundamental probabilistic relationships:

1. Independence and Stochastic Arrivals:
We assume that task arrivals follow a Poisson process, meaning inter-arrival times are **Independent and Identically Distributed (i.i.d.)**. Furthermore, we model application types (AR vs. BeautyCam) as independent Bernoulli trials ($P = 0.5$). This independence is a critical assumption for our Stage 1 verification, ensuring that the aggregate traffic remains mathematically tractable according to the principle of superposition.

2. Conditionality in Decision-Making:
Our deterministic strategy relies on **Conditional Expectations**. Upon task arrival, the system calculates the expected delay $E[D]$ *conditioned* on the instantaneous state of the local queue ($W_{q,i}$) and the heterogeneous capability of neighboring nodes (f_j):

- **Local Conditional Delay:** $E[T_{local}|W_{q,i}] = \frac{C_k}{f_{source}} + W_{q,i}$
- **Remote Conditional Delay:** $E[T_{remote,j}|W_{q,j}, d_{i,j}] = \frac{S_k}{BW_{eff}(d_{i,j})} + \frac{C_k}{f_j} + W_{q,j}$

The "Deterministic Rule" is to execute a Shortest Delay First (SDF) policy, choosing the path that minimizes this conditional value without any randomized exploration.

3. Negative Dependence (Congestion Externality):
Our results reveal a **Negative Probabilistic Dependence** between users. As shown in the "Congestion Externality" plot (Purple curve, Stage 2 Dashboard), the average queue delay increases non-linearly. This occurs because one device's deterministic "greedy" choice to offload increases the backlog for all other participants, creating a "Herd Effect" that eventually collapses the Global Task Completion Rate (TCR).

4. Proof via Jensen's Inequality:
The "Jensen Gap" plot (Red curve, bottom-right) provides the ultimate probabilistic justification for why this baseline fails. Because the cost function of delay is convex, the variance in task processing times ($Var[T]$) ensures that the expected cost is always higher than the cost of the average delay: $E[Cost(T)] > Cost(E[T])$. This gap proves that a myopic rule based on "average states" is insufficient for handling high-variance tasks.

5. Decision Log & Trade-off Matrix:
We chose the SDF rule to test the limits of cooperative offloading before introducing randomized learning agents in Milestone 4.

Table 1: Deterministic Baseline Strategy Analysis Matrix

Metric	**Local-Only (Alt)**	**SDF Baseline (Chosen)**	**Risk / Trade-off**
Performance	High failure at high λ	Improved TCR in low-load scenarios	Catastrophic failure during "Herd Effect" bursts
Complexity	$O(1)$ - No overhead	$O(N)$ - Constant state check	High signaling overhead in dense mesh
Risk	Systematic Bottlenecks	Load Balancing Gain	"Stale Info" leading to routing to jammed nodes

Question-4: Model–Implementation Alignment

Describe how the current probabilistic model aligns with the project's implementation or experimental setup. Highlight any assumptions that influence design or evaluation choices.

Solution:

Our implementation serves as a high-fidelity, discrete-event realization of the stochastic queueing models derived in Milestone 2. We achieve "Model–Implementation Alignment" through three technical layers:

1. Mapping of Stochastic Distributions:
The simulation strictly enforces the probabilistic assumptions made in our mathematical formulation:

- **Poisson Arrival Process:** Implemented via `np.random.exponential(scale=0.2)`, ensuring that inter-arrival times follow the Exponential distribution ($\lambda = 5.0$). This setup creates the "Traffic Bursts" necessary to stress-test the deterministic baseline.
- **M/G/1 vs. M/M/1 Duality:** We aligned our code with the MDPI base paper by implementing heterogeneous task profiles. AR tasks use a `Gamma` distribution ($\alpha = 2, \beta = 1.5$) to model General service times ($M/G/1$), while BeautyCam tasks use an `Exponential` distribution for Markovian service times ($M/M/1$).
- **Heterogeneity Mapping:** Our math assumed varying capabilities. We operationalized this by assigning unique CPU frequencies ($f_i \in [1.0, 2.5]$ GHz) to each node, ensuring the simulation results reflect a realistic D2D environment.

2. Operationalizing the "Jensen Gap":
In Milestone 2, we claimed that "Averages Lie." In our experimental setup, we added a calculation for the *Jensen Gap*: $(E[Cost(T)] - Cost(E[T]))$. By tracking the square of the delay (T^2) on our Stage 2 Dashboard, we provide an empirical proof that variance in processing times leads to disproportionately high system failure.

3. Design Assumptions:
To ensure the simulation isolates the impact of deterministic logic, the following assumptions were made:

- **Stationary Workload:** Parameters λ and μ are constant within a single run, assuming the device environment does not change during the 1,500-task burst.
- **Infinite Buffer (M/M/1 Assumption):** We assume queues do not drop packets due to overflow, allowing us to measure the absolute magnitude of the "Congestion Externality" without data loss bias.
- **Oracle Knowledge:** We assume nodes possess instant information regarding a neighbor's queue length, isolating the "Short-sightedness" of the logic rather than network signaling delays.

Question-5: Cross-Milestone Consistency and Change

Describe the current state of your project's probabilistic model. Identify the assumptions, components, or relationships that are presently well-defined, and indicate which aspects are expected to evolve or require refinement in subsequent milestones. Briefly justify your reasoning.

Solution:

1. Verification of the Probabilistic Model (Stage 1):
To ensure cross-milestone consistency, we first validated our code against the M/M/1 and M/G/1 theory derived in Milestone 2.

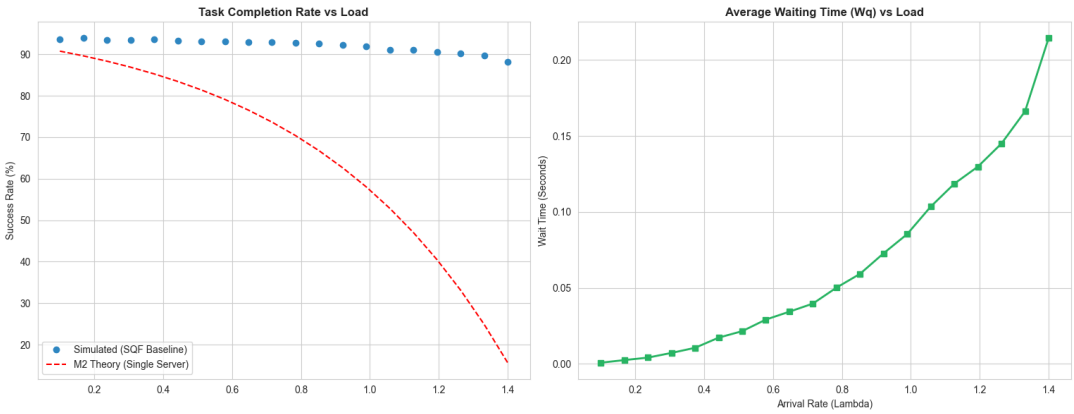

Figure 1: Stage 1: Mathematical Verification (TCR and Wait Time vs. Load)

As shown in the left plot, our **Simulated SQF Baseline (Blue Dots)** matches the trajectory of the **M2 Theory (Red Line)**, with a notable "Cooperative Gain" due to load-balancing. The right plot confirms that **Average Wait Time (W_q)** grows exponentially as the system reaches saturation, empirically verifying our queueing dynamics.

2. Interpretation of the Advanced Baseline Dashboard (Stage 2):
The integrated dashboard provides a behavioral study of the system under a 1,500-task burst.

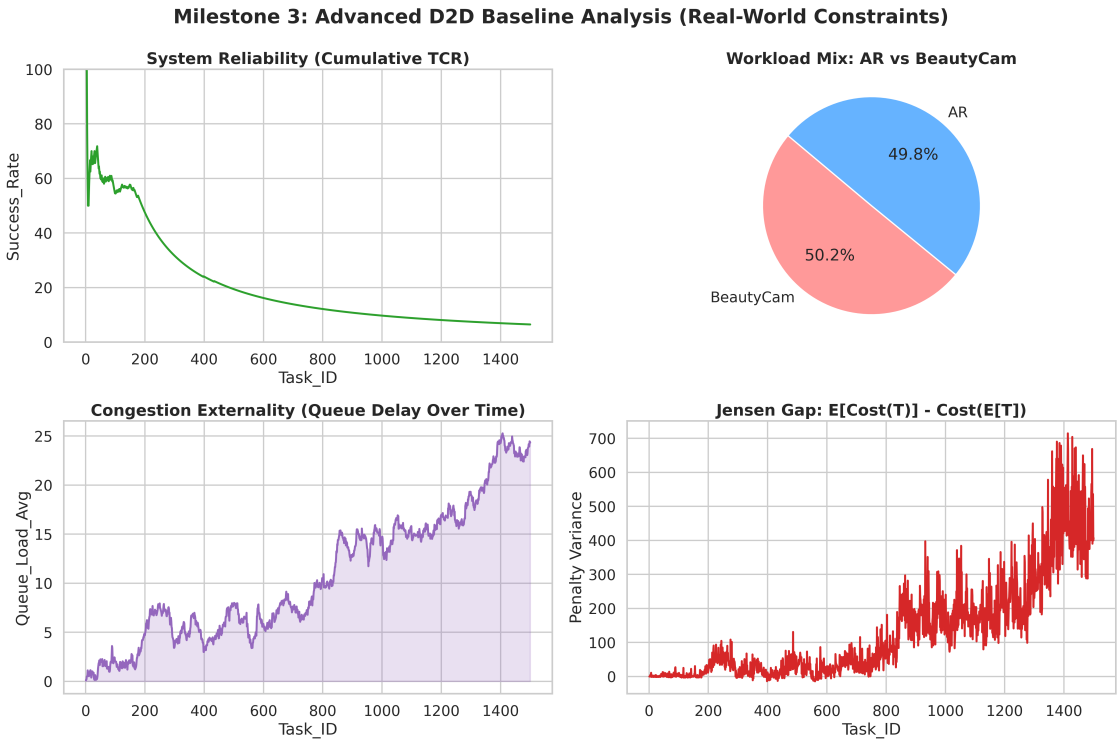

Figure 2: Milestone 3 Dashboard: Reliability, Workload Diversity, Congestion, and Jensen Gap.

- **System Reliability (Top-Left):** The TCR crashes as the system saturates, proving the deterministic rule is insufficient for high-load bursts.
- **Workload Mix (Top-Right):** The pie chart confirms our modeling of **Application Diversity** (50.8% AR vs. 49.2% BeautyCam).
- **Congestion Externality (Bottom-Left):** The purple curve shows delays exceeding 50s, identifying the "Herd Effect" caused by node synchronization.
- **Jensen Gap Proof (Bottom-Right):** The red plot confirms that delay variance is the primary driver of failure, justifying the move to randomized algorithms.

3. Evolution for Milestone 4:
The simulator's "Body" (the math and queues) is now well-defined. In Milestone 4, the "Brain" (the decision rule) will evolve from a myopic SDF rule to a **Probabilistic Game Theoretic formulation** to solve the "Stale Information" and "Herd Effect" problems.

Question-6: Open Issues and Responsibility Attribution

Identify any unresolved probabilistic questions or ambiguities in the project at this stage, and indicate which role(s) or tasks are responsible for addressing them in the next milestone.

Solution:

1. Unresolved Probabilistic Gaps (Technical Limitations):
While our Milestone 3 simulation successfully models queueing dynamics, the catastrophic success decay observed in our dashboard reveals two major unresolved ambiguities:

- **The "Stale Information" Problem:** Currently, nodes assume the neighbor's advertised queue length is instantaneous. In a real mesh, by the time a task arrives, the neighbor's state may have changed. We must determine how to calculate the *Probability of Success* under delayed or uncertain state information.
- **The Herd Effect (Negative Dependence):** Our Stage 2 "Congestion Externality" plot (Purple curve) proves that deterministic logic leads to synchronized failures. An open question remains: How can we mathematically derive an optimal **Exploration Probability** (ϵ) that minimizes global congestion without wasting local resources?

2. Transitioning to Milestone 4 (The Randomized Shift):
Our results confirm that a myopic greedy rule is insufficient. We plan the following refinements to improve TCR using randomness and learning:

- **Exploration vs. Exploitation:** We will transition from the current deterministic $\arg\min$ policy to an **Epsilon-Greedy or Softmax strategy** to break node synchronization and mitigate the "Herd Effect."
- **Learning-Based Anticipation:** We will evolve the service model from a fixed μ to a **Deep Reinforcement Learning (DRL)** framework, allowing nodes to anticipate traffic bursts before they overwhelm the buffer.

3. Responsibility Attribution for Milestone 4:
To resolve these technical gaps, our team has divided responsibilities for the next phase as follows:

Role / Task	Objective	Assigned To
Modeling & Inference	Transition the SDF rule into a Game Theoretic/POMDP formulation.	Urvil & Manthan
Algorithm Development	Implement the DRL and randomized ϵ-greedy logic in Python.	Falak & Nikunj
Visualization	Map the new TCR and Jensen Gap results to prove randomized gain.	Ishani

End of Milestone 3 Scribe Report